



MECHANICAL, ELECTRICAL AND PLUMBING PROJECTS

MEP SYSTEMS

Current Conditions

- Outdated Equipment and Technology
- Past Life Expectancy
- Increasing Operational and Maintenance Cost
- Unreliable Systems
- Lack of Redundancy

MEP SYSTEMS

Goals

- Replace Outdated and Failing Equipment
- New Technology
- Reliability
- Redundant Systems
- Increase Operational Efficiency
- Staff and Inmate Training

MEP SYSTEMS

Project Implementation

- 207 MEP projects will be funded and issued for bid proposal over the next 4 fiscal years.
- These projects are estimated at a total cost of approximately \$660 m.
- Projects will be issued for bid by groups, based on type of work required and geographic locations.
- Projects to utilize inmate labor, where appropriate.
- Fiscal Responsibility to ensure "Best Value" to the Government.

Projects

South Central Region
FCI La Tuna



Projects

South Central Region FCI La Tuna

- FCI La Tuna was constructed in 1932 consists of 25 buildings and approximately 80,185 s.f. under roof.
- Currently does not have a chilled water system.
- Install a complete Chilled Water System: Chillers, Cooling Tower, Pumping Systems and associated piping.
- Install new HVAC system: Roof Top Units, Make-Up Air Units Automated Environmental Controls, Exhaust Systems and all associated ductwork/piping.
- Upgrade the existing Electrical Systems to accommodate new power demand.
- A/E to provide full construction documents and specifications.

Projects

South Central Region
FCI Seagoville



Projects

South Central Region FCI Seagoville

- FCI Seagoville was constructed in 1937, comprised of 38 building with 500,314 s.f. under roof.
- Inmate housing units 5, 7, 10 & 12 (57,746 sq ft) currently have heating and ventilation systems only.
- Heating is accomplished with steam-fired radiators located throughout each building with steam delivered by boilers located in the Powerhouse.
- Currently there are no chillers located within the Powerhouse.
- Provide a complete HVAC system for the 4 buildings to include: Chiller(s), Air Handling Units, Automated Control System, ductwork, piping and pumping systems.
- A/E to provide full construction documents and specifications.
- Historic Preservation concerns.

Projects

South Central Region
FCI Three Rivers



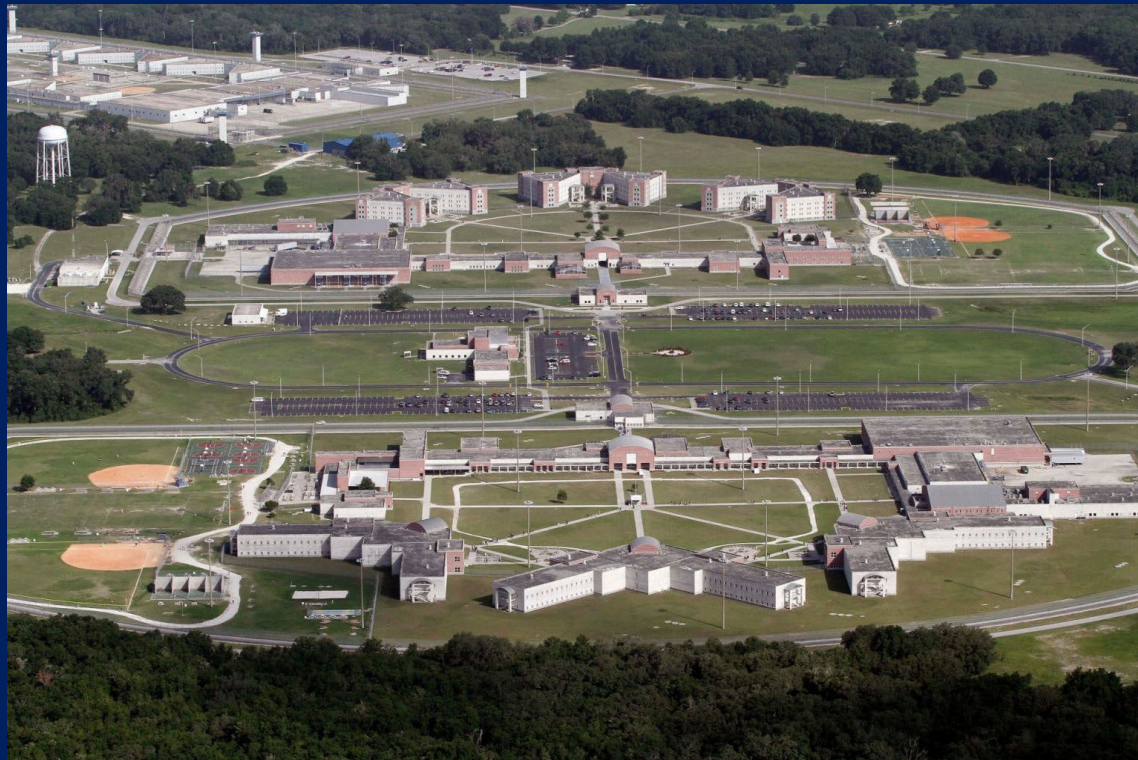
Projects

South Central Region FCI Three Rivers

- FCI Three Rivers was constructed in 1990 and occupies approximately 300 acres.
- Replace two existing 450 kw generators, associated transfer switches and cabling located within the equipment room.
- Replace underground high voltage cabling in feed loop "B".
- Replace leaking high voltage transformers.
- Replace main switch gear and distribution panels in the Food Service Buildings at FCI and Camp.
- A/E to provide full construction documents and specifications.

Projects

South Central Region
FCC Beaumont



Projects

South Central Region FCC Beaumont

- FCC Beaumont was constructed from 1996 to 1998 , consists of one USP, Two FCIs and one Satellite Prison Camp totaling 1,641,623 s.f. and situated on 960 acres.
- Existing complex sewer lift station is in a low-lying area subject to ground infiltration.
- Install new sewer lift station with proper elevations to ensure compliant operation.
- Existing underground domestic and fire water piping, all associated valves and pumping systems are severely corroded due to the corrosive environment and past flooding/storms
- Replace all underground piping system to include isolation valves.
- Replace all damaged piping in the Remote Utility Buildings (3) to include fire/domestic water systems and associated controls.
- A/E to provide full construction documents and specifications.

Projects

South Central Region
FMC Fort Worth



Projects

South Central Region FMC Fort Worth

- FMC Fort Worth was constructed in 1934 consists of 31 buildings and occupies approximately 38 acres.
- Construct one domestic water storage tank to hold one million gallons of water.
- Addition of this tank will allow for 10 days of water to be used during emergency events.
- A/E to provide full construction documents and specifications.
- Historic Preservation concerns.

Water Storage Tanks Projects

- Northeast Region – FCI Berlin, USP Allenwood, USP Lewisburg
- Mid-Atlantic Region – FCI Ashland, FCI Petersburg, FCI Morgantown, FCI Cumberland
- North Central Region – USP Terre Haute, FCI Cumberland
- South Central Region – FCI La Tuna, FCC Beaumont, FCC Oakdale, FCI Texarkana, FCI Carswell
- Western Region – FCI Safford, USP Lompoc

Projects

Western Region
FCI Herlong



Projects

Western Region FCI Herlong

- FCI Herlong was constructed in 2005, consists of 23 building with approximately 560,000 s.f. under roof.
- Replace existing heating hot water distribution loop due to excessive leaks.
- Current loop extends from Central Utility Plant to multiple building entries throughout the complex.

Projects

Western Region

FCC Victorville - FCI-2



Projects

Western Region FCI Victorville

- FCC Victorville was constructed in 2000, consists of one USP, Two Medium FCIs and one Satellite Prison Camp totaling 1,732,420 s.f. and situated on 960 acres.
- The work will be isolated to FCI – 2.
- Replace two centrifugal Chillers and two Cooling Towers and associated piping located with-in the Central Utility Plant.
- Install a heat recovery system to capture rejected heat and utilize it to supplement the hot water boilers.

Projects

Western Region
MCC San Diego



Projects

Western Region MCC San Diego

- MCC San Diego was constructed in 1974, consists of 23 stories and approximately 225,000 sf.
- Replace all domestic water lines throughout the building with selection demolition required.
- Access to some piping will be require demolition of various existing building components to replace.
- Building repair to current conditions will be required.

Projects

Northeast Region
FCI Berlin



Projects

Northeast Region FCI Berlin

- FCI Berlin was constructed in 2010 consists of 689,909 s.f. and situated on approximately 707 acres
- Replace existing pre-insulated heating hot water distribution loop due significant leaks.
- Current loop extends from Central Utility Plant to multiple building entries throughout the complex.

Projects

Northeast Region
FCI Schuylkill



Projects

Northeast Region FCI Schuylkill

- FCI Schuylkill was constructed in 1991 consists of 516,397 s.f. and situated on approximately 600 acres
- Replace existing pre-insulated heating hot water distribution loop due to significant leaks.
- Current loop extends from Central Utility Plant to multiple building entries throughout the complex.

Projects

Southeast Region FCC Coleman



Projects

Southeast Region USP 1 / USP 2 Coleman

- FCC Coleman was constructed in 1993, consists of two USPs, Two FCIs and one Prison Camp totaling 2,434,578 s.f. and situated on 1,746 acres.
- Replace two centrifugal Chillers and associated pumping systems located with-in the Central Utility Plant at USP-2.
- Replace Automated Controls Systems and Repair/Replace Boilers and associated pumping systems at USP-1.

Projects

Southeast Region
FCI Atlanta



Projects

Southeast Region FCI Atlanta

- FCI Atlanta was constructed in 1902 consists of 47 buildings totaling approximately 1.6 million sq.ft.
- Currently the vast majority of the electrical distribution system is in need of major repair or replacement.
- Repair existing basement and subsurface drainage system to eliminate heavy water intrusion.
- Generator #3, all paralleling switchgear and automatic transfer switches are in need of replacement.
- Upgrade all existing electrical switchgear, distribution panels and transformers throughout the institutional grounds.
- Replace all High Mast Light Poles.
- A/E to provide full construction documents and specifications.
- Historic Preservation concerns.

Projects

Mid-Atlantic Region
FCI Morgantown



Projects

Mid-Atlantic Region FCI Morgantown

- FCI Morgantown was constructed in 1968, consists 22 buildings totaling 369,000 s.f. and occupies 400 acres.
- Currently the institution has no means of supplying emergency power in the event of commercial outage.
- Existing main electrical switch gear and distribution panels are overloaded and/or in need or replacement
- Install two new generators with switchgear, fuel tanks, emergency panels and all underground cabling required to adequately provide reliable power during times of emergency or inclement weather conditions.
- Upgrade all existing electrical switchgear, distribution panels and transformers throughout the institutional grounds.
- A/E to provide full construction documents and specifications.

Projects

Mid-Atlantic Region
FMC Lexington



Projects

Mid-Atlantic Region FMC Lexington

- FMC Lexington was constructed in 1931 is comprised of 34 buildings with a total of 1,051,096 square feet.
- Currently the institution has no means of supplying emergency power in the event of commercial outage.
- Existing main electrical switch gear and distribution panels are overloaded and/or in need or replacement.
- Install two new generators with switchgear, fuel tanks, emergency panels and all underground cabling required to adequately provide reliable power during times of emergency or inclement weather conditions.
- Upgrade all existing electrical switchgear, distribution panels and transformers throughout the institutional grounds.
- A/E to provide full construction documents and specifications.
- Historic Preservation concerns.

Projects

Mid-Atlantic Region
FCI Ashland



Projects

Mid-Atlantic Region FCI Ashland

- FCI Ashland was constructed in 1940 is comprised of approximately 500,000 and encompasses 143 acres.
- Currently the institution domestic water needs are being service a 5" bronze water main located within the utility tunnel. Main and all laterals have signs of significant deterioration.
- Current pipe sizing is not adequate to meet the demands.
- Replace the entire domestic water system located throughout the institution grounds to include piping, valves and backflow prevention devices.
- A/E to provide full construction documents and specifications.
- Historic Preservation concerns.

Projects

North Central Region
FCI Englewood



Projects

North Central Region

FCI Englewood

- FCI Englewood was constructed in 1938 is comprised of 15 buildings with a total of 263,000 square feet
- Existing electrical infrastructure is up to 35 years old and past it's serviceable life cycle.
- 13.2KV switchgear located in the Powerhouse (frequent operational failures)
- Main and feeder relays(existing GE 750 relays obsolete, upgrade to digital relays)
- Replace main motor control center in Powerhouse (reliability and overheating issues)
- Replace underground feeder throughout the institution (installed in 1938, loss of integrity, undersized)
- Upgrade medium voltage, distribution panels and transformers throughout the institutional grounds.
- A/E to provide full construction documents and specifications.
- Historic Preservation concerns.

Projects

North Central Region
FCI Marion



Projects

North Central Region

FCI Marion

- FCI Marion was constructed in 1963 is comprised of 15 buildings and occupies 935 acres
- Existing electrical infrastructure is up to 50 years old and past it's serviceable life cycle.
- Replace six medium voltage pad mount transformers.
- Replace electrical substations located within five different transformer vaults.
- Replace current grounding system (electronic equipment and staff safety issues.)
- Replace high voltage underground feeders throughout the FCI
- Replace substation and distribution feeders and panel at FPC
- A/E to provide full construction documents and specifications.